

3. All mammals, including humans, are examples of holozoic heterotrophs. Once food has been ingested, it undergoes both mechanical and chemical digestion.

(a) Describe what is meant by the terms mechanical digestion and chemical digestion. [2]

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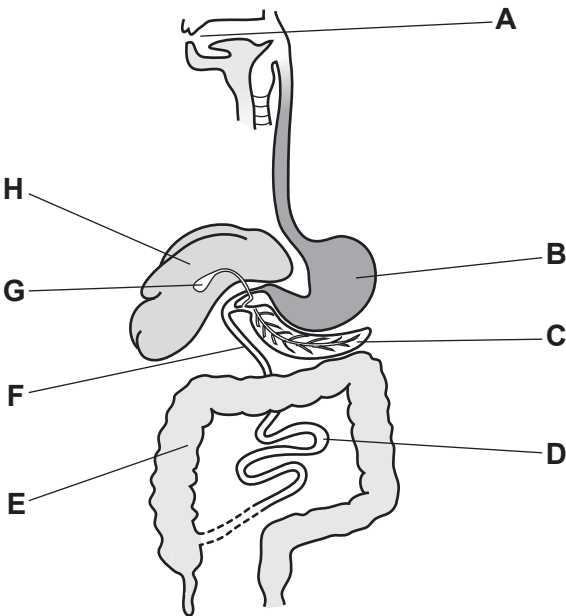
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(b) **Image 3.1** shows the human alimentary canal and associated organs.

Image 3.1



(i) Using the letters from **Image 3.1**, identify the organs where lipase and bile are synthesised. [1]

Lipase:

Bile:

(ii) Bile is composed of bile salts and HCO_3^- ions. Explain the importance of the HCO_3^- ions in the small intestine. [2]

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- (c) An experiment was carried out to determine the effect of bile salts on the digestion of lipids. The student used full fat milk as it contains a relatively high concentration of lipids. Each test tube was set up as shown in **Table 3.2**; all solutions were maintained at 37 °C throughout the investigation.

Table 3.2

Tube	Volume / cm ³					
	Full fat milk (3.5% fat content)	Sodium carbonate solution (0.05 mol dm ⁻³)	Lipase solution (5%)	Bile salts solution (3%)	Phenolphthalein (1%)	Distilled water
A	5.0	1.0	0.5	0.0	0.2	1.0
B	5.0	1.0	0.5	1.0	0.2	0.0
C	5.0	1.0	0.0	1.0	0.2	0.5

Phenolphthalein was used as an indicator. It is pink in solutions with a pH above 10 and colourless in solutions below pH 8.3.

- (i) Explain why it was important that sodium carbonate solution was added to each of the test tubes. [1]

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- (ii) Explain why it was important that different volumes of water were added to each of the test tubes. [1]

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- (d) The student recorded the time taken for the phenolphthalein to turn colourless. The results are shown in **Table 3.3**.

Table 3.3

Tube	Time taken for phenolphthalein to turn colourless/s			
	Trial 1	Trial 2	Trial 3	Mean
A	247	283	266	265
B	111	87	109	102
C	*	*	*	*

* No colour change was recorded in test tube **C** even after 900 seconds.



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(i) Explain the results shown in test tubes **A** and **B**. [4]

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(ii) Explain why bile salts were included in test tube **C** but lipase was not. [1]

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(iii) Suggest **one** source of inaccuracy with this investigation and suggest **one** possible way in which it could be improved. [2]

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